Duty at home

DM8 · A complete sequence · 60 min.

For every $n \in \mathbb{N}$, we're laying $I_n = \int_0^1 x^n e^x \, \mathrm{d}x$. For $n = 0$, integrande is the function $x \mapsto e^x$.

1. Calculate I_0then I_1 with the help of party integration.
2. Demonstrate that, for everything $n \in \mathbb{N}$, $I_{n+1} = e - (n+1)I_n$.
3. By comparing the integrands, demonstrate that (I_n) is positive and decreasing.
4. Show that $\frac{1}{n+1} \leq I_n \leq \frac{e}{n+1}$ for every $n \in \mathbb{N}$. Deduct the limit of (I_n).
5. In the calculation of I_{n+1} by the relationship of question 2, by what factor an error on I_n Is she If the other errors are neglected, explain the risk of a repeated calculation of the that way.